Code No: 155AB **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August - 2022
ADVANCED OPERATING SYSTEMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) What is distributed operating system and write its characteristics and advantages.
b) Describe blocking and non blocking communication primitives in distributed operating system. [7+8]

2.a) Explain frame work of logical clocks.
b) Give the architecture of distributed operating system. [8+7]

3.a) Define critical section problem and explain with an example.
b) State and explain Singhal's heuristic algorithm for Mutual exclusion. [7+8]

4.a) Write Lamport's algorithm for distributed mutual exclusion.
b) Device performance and optimality for Lamport's algorithm for distributed mutual exclusion. [7+8]

5.a) Describe deadlock handling strategies in distributed operating system.
b) Explain edge chasing algorithm for deadlock detection. [7+8]

6.a) Explain wait-for graph with an example.
b) Discuss diffusion computing algorithm for dead lock detection. [7+8]

7.a) Explain the architecture of distributed file system.
b) Discuss the issues in multi processor operating systems and compare with distributed operating system. [7+8]

8.a) Elaborate components of load distribution algorithm.
b) Write the advantages and abstract view of distributed shared memory. [7+8]

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